

Lab: Planning a Color Palette

Materials Exploration — Planning

Objective

Practice the full planning cycle for a crochet project, with special focus on color palette design, yarn selection, and yardage estimation. You will experience planning as Active Inference — generating predictions, testing them, and revising.

Materials Needed

- Access to a yarn stash or yarn shop (physical or online)
- A simple pattern (e.g., a striped scarf, granny square blanket, or color-block cowl)
- A color wheel (printed or digital)
- A calculator
- A notebook for planning
- Optional: an inspiration image (a photograph, painting, or natural scene with appealing colors)

Part 1: Inspiration to Palette (20 minutes)

1. Choose an inspiration source — a photograph, a painting, a landscape, or even a favorite outfit.
2. Identify 3-5 colors in the inspiration that you would like to translate into yarn.
3. Using the color wheel, classify your chosen colors:
4. Are they warm or cool?
5. Are they analogous, complementary, or triadic?
6. What is the contrast level (high, medium, low)?
7. Search your stash or a yarn shop for yarns that approximate these colors.
8. Lay the yarn samples together. Does the palette match your vision?

Reflection: You have just performed expected free energy minimization — simulating the future visual outcome and selecting materials to minimize the gap between vision and reality.

Part 2: Project-Yarn Matching (15 minutes)

1. Choose a simple pattern.
2. For each candidate yarn, evaluate:
3. Is the weight appropriate? (Check gauge)
4. Is the fiber suitable? (Function, care, season)
5. Is the texture compatible? (Will stitch pattern show?)
6. Is the budget reasonable?
7. Rank your candidates. Which best matches the project?

Part 3: Yardage Planning (15 minutes)

1. Using the pattern's yardage requirements, calculate how much of each color you need.
2. Add a 15% safety margin.
3. Check: Are all your yarns from the same dye lot? If not, plan for how to manage the difference.
4. Calculate the total cost.

Color	Yarn	Yards Needed	+ 15% Safety	Skeins to Buy	Dye Lot?	Cost

Part 4: Swatch and Revise (15 minutes)

1. Crochet a small test swatch using your planned palette.
2. Evaluate:
3. Do the colors work together in fabric as well as in the skein?
4. Is the gauge correct?
5. Does the stitch pattern show well in these yarns?
6. Revise your plan based on the swatch results. What needs to change?

Reflection: The swatch is the planning test — it generates evidence that either confirms or contradicts your plan. How many revisions did your plan need?

Discussion Questions

1. How did your palette change from inspiration to final yarn selection? What compromises were necessary?
2. Was yardage estimation straightforward, or did it require significant uncertainty management?
3. What was the most challenging aspect of planning — color selection, yarn matching, or yardage math?

Reflection

Write a short paragraph about planning in the fiber arts. How does the planning process connect to everything you have learned in this course — systems, agents, perception, cognition, action, learning, and communication? How does planning bring it all together?